Math for EE

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ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9–12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

1. GATE BM 2025

1. GATE BM 2025

1. The 12 musical notes are given as $C, C^{\#}, D, D^{\#}, E, F, F^{\#}, G, G^{\#}, A, A^{\#}, B$. Frequency of each note is $\sqrt[12]{2}$ times the frequency of the previous note. If the frequency of the note C is 130.8 Hz, then the ratio of frequencies of notes $F^{\#}$ and C is: (GATE BM 2025)

   (a) $\sqrt[6]{2}$
   (b) $\sqrt{2}$
   (c) $\sqrt[4]{2}$
   (d) 2

   Solution:

   Each successive musical note has a frequency $\sqrt[12]{2}$ times that of the previous note.

   From C to $F^{\#}$, there are 6 successive intervals:

$$C \to C^{\#} \to D \to D^{\#} \to E \to F \to F^{\#} \tag{1.1}$$

   Therefore,

$$\frac{f_{F\#}}{f_C} = \left(\sqrt[12]{2}\right)^6 \tag{1.2}$$

$$= 2^{6/12} = 2^{1/2} = \sqrt{2} \tag{1.3}$$

   Hence,

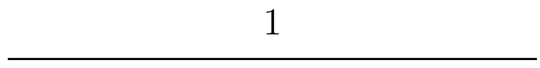
$$\boxed{\frac{f_{F\#}}{f_C} = \sqrt{2}} \tag{1.4}$$

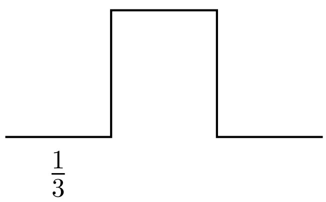
   Therefore, the correct answer is

$$\boxed{\text{(b) } \sqrt{2}} \tag{1.5}$$

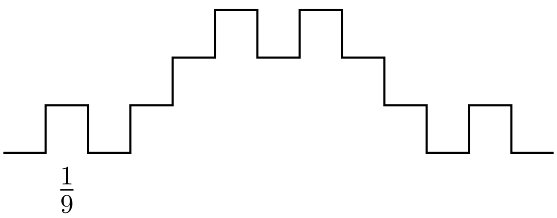
2. The following figures show three curves generated using an iterative algorithm. The total length of the curve generated after "Iteration n" is:

GATE BM 2025

Iteration 0: 

Iteration 1: 

Length of each segment: $\frac{1}{3}$

Iteration 2: 

Length of each segment: $\frac{1}{9}$

Note: The figures shown are representative.

(a) $\left(\frac{5}{3}\right)^{\frac{n}{2}}$
(b) $\left(\frac{5}{3}\right)^{n}$
(c) $\left(\frac{5}{3}\right)^{2n}$
(d) $\left(\frac{5}{3}\right)^{n(2n-1)}$

Solution:

At iteration 0, the length of the curve is 1.

$$L_0 = 1 \tag{2.1}$$

At iteration 1, the curve consists of 5 segments, each of length $\frac{1}{3}$.

Therefore,

$$L_1 = 5\left(\frac{1}{3}\right) = \frac{5}{3} \tag{2.2}$$

At every subsequent iteration, each segment is replaced by 5 segments, each having $\frac{1}{3}$ of the length of the original segment. Hence, the total length is multiplied by

$$\frac{5}{3} \tag{2.3}$$

Therefore, after n iterations,

$$L_n = L_0\left(\frac{5}{3}\right)^n \tag{2.4}$$

Since $L_0 = 1$,

$$L_n = \left(\frac{5}{3}\right)^n \tag{2.5}$$

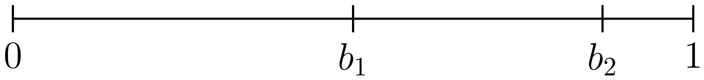
Hence, the correct answer is

$$\boxed{\text{(b)} \ \left(\frac{5}{3}\right)^n} \tag{2.6}$$

3. A stick of length one meter is broken at two locations at distances of b_1 and b_2 from the origin (0), as shown in the figure. Note that $0 < b_1 < b_2 < 1$. Which one of the following is NOT a necessary condition for forming a triangle using the three pieces?

   Note: All lengths are in meter. The figure shown is representative.

   (GATE BM 2025)



(A) $b_1 < 0.5$

(B) $b_2 > 0.5$

(C) $b_2 < b_1 + 0.5$

(D) $b_1 + b_2 < 1$

Solution:

The stick of length 1 m is broken at b_1 and b_2.

Therefore, the lengths of the three pieces are

$$l_1 = b_1, \qquad l_2 = b_2 - b_1, \qquad l_3 = 1 - b_2. \tag{3.1}$$

For three lengths to form a triangle, the sum of any two sides must be greater than the third side.

Therefore,

$$l_1 + l_2 > l_3 \tag{3.2}$$

$$b_1 + (b_2 - b_1) > 1 - b_2 \tag{3.3}$$

$$b_2 > 1 - b_2 \tag{3.4}$$

$$b_2 > \frac{1}{2}. \tag{3.5}$$

Hence, option (B) is a necessary condition.

Now consider

$$l_1 + l_3 > l_2. \tag{3.6}$$

Substituting the lengths,

$$b_1 + (1 - b_2) > b_2 - b_1. \tag{3.7}$$

Therefore,

$$2b_1 + 1 > 2b_2 \tag{3.8}$$

or

$$b_2 < b_1 + \frac{1}{2}. \tag{3.9}$$

Hence, option (C) is also a necessary condition.
Finally,

$$l_2 + l_3 > l_1. \tag{3.10}$$

Substituting,

$$(b_2 - b_1) + (1 - b_2) > b_1. \tag{3.11}$$

Therefore,

$$1 - b_1 > b_1 \tag{3.12}$$

which gives

$$b_1 < \frac{1}{2}. \tag{3.13}$$

Hence, option (A) is also a necessary condition.

However, the condition

$$b_1 + b_2 < 1 \tag{3.14}$$

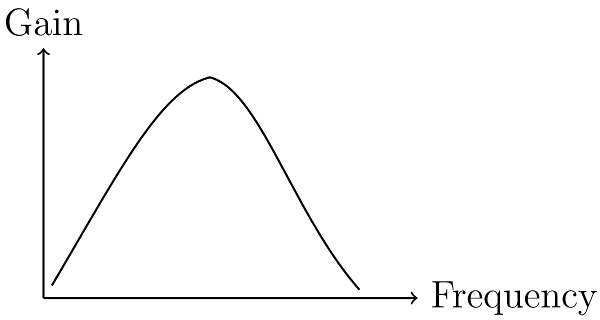
is not required for the three pieces to form a triangle.

Therefore, the condition which is **NOT** necessary is

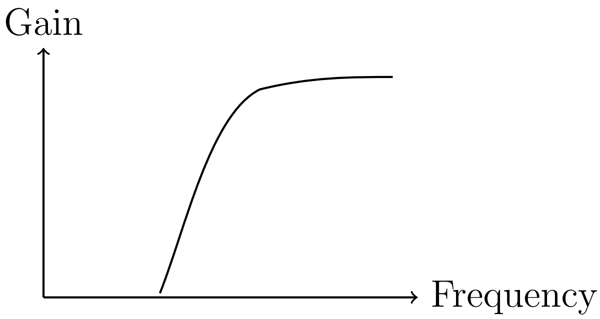
$$\boxed{\text{(D)} \quad b_1 + b_2 < 1} \tag{3.15}$$

4. Which one of the following represents the frequency response of a notch filter?

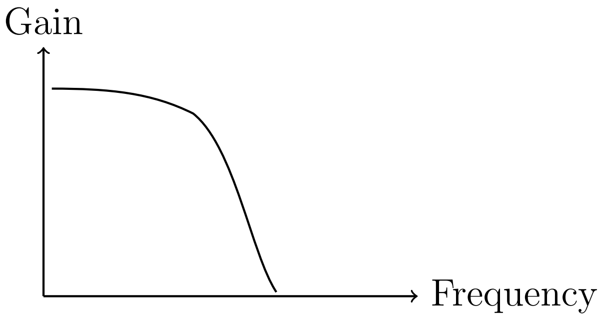
(GATE BM 2025)



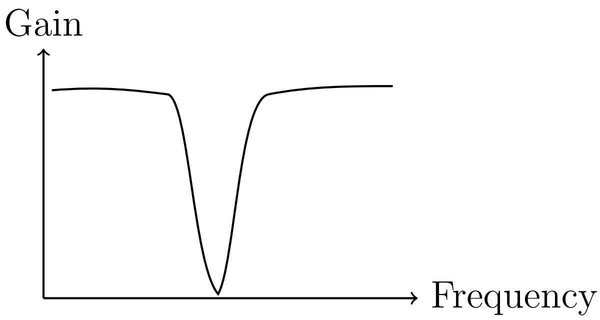
(A)



(B)



(C)

(D) 

Solution:

A notch filter is a band-stop filter that rejects or strongly attenuates a narrow range of frequencies while allowing frequencies outside that range to pass.

Therefore, its frequency response has high gain at low and high frequencies, with a sharp reduction in gain at a particular frequency.

Mathematically, if ω_0 is the rejected frequency, then the response has a deep attenuation around ω_0:

$$|H(j\omega_0)| \approx 0 \tag{4.1}$$

while frequencies away from ω_0 are passed with relatively high gain.

$$|H(j\omega)| \text{ is high for } \omega < \omega_0 \text{ and } \omega > \omega_0. \tag{4.2}$$

Among the given frequency responses, only option (D) shows a sharp reduction in gain at a particular frequency while allowing frequencies on either side to pass.

$$Hence, \boxed{\text{Answer: (D)}} \tag{4.3}$$

5. For any continuous and differentiable function $f(x)$ with first derivative $f'(x) \neq 0$ for all values of x, which one of the following is ALWAYS TRUE for $a \neq b$?

   Here, a and b are finite and real.

   (GATE BM 2025)

(A) $f(a) > f(b)$

(B) $f(a) = f(b)$

(C) $f(a) \neq f(b)$

(D) $f(a) < f(b)$

Solution:

We are given that $f(x)$ is continuous and differentiable, and

$$f'(x) \neq 0 \qquad \text{for all } x. \tag{5.1}$$

Suppose, for contradiction, that

$$f(a) = f(b). \tag{5.2}$$

Since $f(x)$ is continuous on $[a, b]$ and differentiable on (a, b), Rolle's theorem can be applied.

Rolle's theorem states that if

$$f(a) = f(b), \tag{5.3}$$

then there must exist at least one point $c \in (a, b)$ such that

$$f'(c) = 0. \tag{5.4}$$

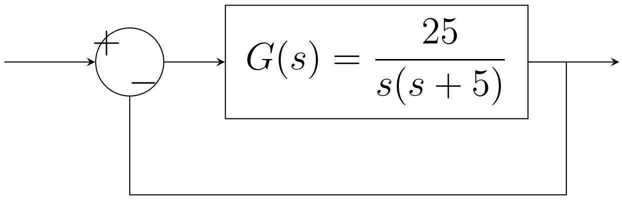
But this contradicts the given condition

$$f'(x) \neq 0 \qquad \text{for all } x. \tag{5.5}$$

Therefore, our assumption that $f(a) = f(b)$ is false. Hence,

$$\boxed{f(a) \neq f(b)}. \tag{5.6}$$

6. The closed loop system shown below ________.



(GATE BM 2025)

(A) is overdamped

(B) is underdamped

(C) is critically damped

(D) has sustained oscillations

Solution:

The given forward transfer function is

$$G(s) = \frac{25}{s(s+5)}. \tag{6.1}$$

Since the feedback is unity and negative, the closed-loop transfer function is obtained using the feedback relation

$$\frac{C(s)}{R(s)} = \frac{G(s)}{1+G(s)}. \tag{6.2}$$

Substituting $G(s)$ from (6.1),

$$\frac{C(s)}{R(s)} = \frac{\dfrac{25}{s(s+5)}}{1+\dfrac{25}{s(s+5)}}. \tag{6.3}$$

Simplifying,

$$\frac{C(s)}{R(s)} = \frac{25}{s(s+5)+25}. \tag{6.4}$$

Therefore,

$$\boxed{\frac{C(s)}{R(s)} = \frac{25}{s^2+5s+25}} \tag{6.5}$$

For a unit-step input,

$$R(s) = \frac{1}{s}. \tag{6.6}$$

Hence,

$$C(s) = \frac{25}{s(s^2+5s+25)}. \tag{6.7}$$

The characteristic equation of the closed-loop system is

$$s^2 + 5s + 25 = 0. \tag{6.8}$$

Comparing this with the standard second-order form

$$s^2 + 2\zeta\omega_n s + \omega_n^2 = 0, \tag{6.9}$$

we obtain

$$\omega_n^2 = 25 \quad \Rightarrow \quad \omega_n = 5, \tag{6.10}$$

and

$$2\zeta\omega_n = 5. \tag{6.11}$$

Therefore,

$$2\zeta(5) = 5 \tag{1}$$

$$\zeta = \frac{1}{2}. \tag{6.12}$$

Since

$$0 < \zeta < 1, \tag{6.13}$$

the system is underdamped.

Thus, the correct answer is

$$\boxed{\text{(B) is underdamped}} \tag{6.14}$$

7. The eigenvalues of the matrix

$$\begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix} \tag{2}$$

are __________.

(GATE BM 2025)

(A) $\cos\theta$ and $\sin\theta$

(B) $e^{i\theta}$ and $e^{-i\theta}$, where $i = \sqrt{-1}$

(C) $\cos\theta$ and $-\sin\theta$

(D) $\cos 2\theta$ and $\sin 2\theta$

Solution:

Let the given matrix be

$$A = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}. \tag{7.1}$$

The eigenvalues λ are obtained from the characteristic equation

$$\det(A - \lambda I) = 0. \tag{7.2}$$

Therefore,

$$\begin{vmatrix} \cos\theta - \lambda & -\sin\theta \\ \sin\theta & \cos\theta - \lambda \end{vmatrix} = 0. \tag{7.3}$$

Expanding the determinant,

$$(\cos\theta - \lambda)^2 + \sin^2\theta = 0. \tag{7.4}$$

Expanding further,

$$\cos^2\theta - 2\lambda\cos\theta + \lambda^2 + \sin^2\theta = 0. \tag{7.5}$$

Using

$$\sin^2\theta + \cos^2\theta = 1, \tag{7.6}$$

we get

$$\lambda^2 - 2\lambda\cos\theta + 1 = 0. \tag{7.7}$$

Using the quadratic formula,

$$\lambda = \frac{2\cos\theta \pm \sqrt{4\cos^2\theta - 4}}{2}. \tag{7.8}$$

Since

$$4\cos^2\theta - 4 = -4\sin^2\theta, \tag{7.9}$$

we have

$$\lambda = \frac{2\cos\theta \pm 2i\sin\theta}{2}. \tag{7.10}$$

Hence,

$$\lambda = \cos\theta \pm i\sin\theta. \tag{7.11}$$

Using Euler's formula,

$$e^{i\theta} = \cos\theta + i\sin\theta, \tag{7.12}$$

and

$$e^{-i\theta} = \cos\theta - i\sin\theta, \tag{7.13}$$

the eigenvalues are

$$\boxed{\lambda_1 = e^{i\theta}, \qquad \lambda_2 = e^{-i\theta}}. \tag{7.14}$$

Therefore, the correct answer is

$$\boxed{\text{(B)}} \tag{7.15}$$

8. If $f(x) = x - \frac{1}{x}$, the value of $\lim_{\Delta x \to 0} \frac{f(1+\Delta x) - f(1)}{\Delta x}$ is ________. (GATE BM 2025)

(A) 0

(B) $\frac{1}{2}$

(C) 1

(D) 2

Solution:

The given limit is the definition of the derivative of $f(x)$ at $x = 1$:

$$f'(1) = \lim_{\Delta x \to 0} \frac{f(1+\Delta x) - f(1)}{\Delta x}. \tag{3}$$

Given

$$f(x) = x - \frac{1}{x}, \tag{4}$$

differentiate with respect to x:

$$f'(x) = 1 + \frac{1}{x^2}. \tag{5}$$

Therefore,

$$f'(1) = 1 + \frac{1}{1^2} = 2. \tag{6}$$

Hence,

$$\boxed{2} \tag{7}$$

Therefore, the correct option is

$$\boxed{\text{(D) } 2}. \tag{8}$$

9. The elements of the dataset $\{-5, 1, a, 5, b\}$ are in ascending order. If both mean and median of the dataset are equal to 3, what is the value of b? (GATE BM 2025)

   (a) 7
   (b) 11
   (c) 9
   (d) 12

Solution:

The dataset consists of 5 elements arranged in ascending order:

$$-5 \leq 1 \leq a \leq 5 \leq b \tag{9.1}$$

Since the dataset has an odd number of elements ($n = 5$), the median is the middle (3rd) element:

$$\text{Median} = a = 3 \tag{9.2}$$

The mean of the dataset is given as 3:

$$\text{Mean} = \frac{-5 + 1 + a + 5 + b}{5} = 3 \tag{9.3}$$

Substituting $a = 3$ into equation (9.3):

$$\frac{-5 + 1 + 3 + 5 + b}{5} = 3 \tag{9.4}$$

$$\frac{4 + b}{5} = 3 \tag{9.5}$$

$$4 + b = 15 \tag{9.6}$$

$$b = 11 \tag{9.7}$$

Hence, the correct answer is

$$\boxed{\text{(b) } 11} \tag{9.8}$$

10. A time domain signal is given by

$$x(t) = 20\sin(100\pi t) + 36\sin(150\pi t) - 2\sin(300\pi t), \tag{9}$$

where time t is in seconds. Among the following options, what is the frequency (in Hz) at which $x(t)$ should be sampled for accurate signal reconstruction? (GATE BM 2025)

(a) 500
(b) 150
(c) 200
(d) 250

Solution:

To avoid aliasing and accurately reconstruct the signal, the sampling frequency f_s must satisfy the Nyquist sampling criterion:

$$f_s \geq 2f_{\max} \tag{10.1}$$

where $f_{\max}$ is the highest frequency component in the signal $x(t)$.

The signal consists of three sinusoidal components with angular frequencies ω_1, ω_2, and ω_3:

$$\omega_1 = 100\pi \implies f_1 = \frac{100\pi}{2\pi} = 50 \text{ Hz} \tag{10.2}$$

$$\omega_2 = 150\pi \implies f_2 = \frac{150\pi}{2\pi} = 75 \text{ Hz} \tag{10.3}$$

$$\omega_3 = 300\pi \implies f_3 = \frac{300\pi}{2\pi} = 150 \text{ Hz} \tag{10.4}$$

The maximum frequency component present in the signal is:

$$f_{\max} = 150 \text{ Hz} \tag{10.5}$$

According to the Nyquist theorem, the minimum required sampling rate (Nyquist rate) is:

$$f_s \geq 2 \times 150 \text{ Hz} = 300 \text{ Hz} \tag{10.6}$$

Among the given options, the only frequency greater than or equal to 300 Hz that guarantees accurate signal reconstruction is 500 Hz.

Hence, the correct answer is

$$\boxed{\text{(a) } 500} \tag{10.7}$$

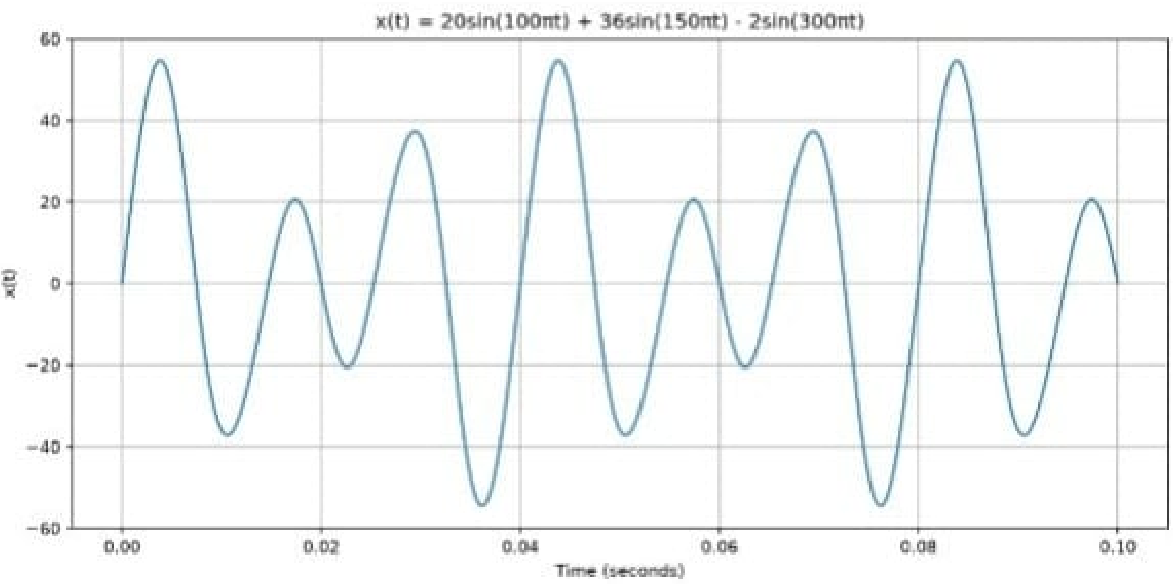


Figure 1: Plot of x(t) vs time

11. A voltage source $V_s = 5\sin\left(100t + \frac{\pi}{2}\right)$ volts is driving a current $I = 10\sin\left(100t + \frac{\pi}{6}\right)$ mA through a circuit. The average power dissipated in the circuit is ________ mW. (GATE BM 2025)

(a) 0.0

(b) 12.5

(c) 25.0

(d) 50.0

Solution:

The formula for the average power P_{avg} dissipated in an AC circuit is given by:

$$P_{\text{avg}} = V_{\text{rms}} I_{\text{rms}} \cos(\theta) \tag{11.1}$$

where:

- $V_{\text{rms}} = \frac{V_m}{\sqrt{2}}$ is the RMS value of the voltage,
- $I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$ is the RMS value of the current,
- $\theta = \phi_v - \phi_i$ is the phase difference between voltage and current.

Given values:

$$V_m = 5 \text{ V}, \quad \phi_v = \frac{\pi}{2} \text{ rad} \tag{11.2}$$

$$I_m = 10 \text{ mA}, \quad \phi_i = \frac{\pi}{6} \text{ rad} \tag{11.3}$$

Calculate the phase difference θ:

$$\theta = \frac{\pi}{2} - \frac{\pi}{6} = \frac{3\pi - \pi}{6} = \frac{2\pi}{6} = \frac{\pi}{3} \text{ rad } (60^\circ) \tag{11.4}$$

Substitute the values into the average power formula:

$$P_{\text{avg}} = \left(\frac{5}{\sqrt{2}}\right) \times \left(\frac{10}{\sqrt{2}}\right) \times \cos\left(\frac{\pi}{3}\right) \tag{11.5}$$

$$P_{\text{avg}} = \frac{50}{2} \times \frac{1}{2} \text{ mW} \tag{11.6}$$

$$P_{\text{avg}} = 25 \times 0.5 = 12.5 \text{ mW} \tag{11.7}$$

Hence, the correct answer is

$$\boxed{\text{(b) } 12.5} \tag{11.8}$$

12. N independent trials of a binomial random variable yield an expectation value of 16 and variance of 12. What is the value of N? (GATE BM 2025)

    (a) 80
    (b) 60
    (c) 45
    (d) 64

Solution:

For a Binomial distribution $X \sim B(N, p)$ with N independent trials and probability of success p:

The expectation value (mean) is given by:

$$E(X) = Np = 16 \tag{12.1}$$

The variance is given by:

11.png

Figure 2: Power vs time

$$\mathrm{Var}(X) = Np(1-p) = 12 \tag{12.2}$$

Divide equation (12.2) by equation (12.1):

$$\frac{Np(1-p)}{Np} = \frac{12}{16} \tag{12.3}$$

$$1 - p = \frac{3}{4} = 0.75 \tag{12.4}$$

Solving for p:

$$p = 1 - 0.75 = 0.25 = \frac{1}{4} \tag{12.5}$$

Now, substitute the value of p back into equation (12.1):

$$N\left(\frac{1}{4}\right) = 16 \tag{12.6}$$

$$N = 16 \times 4 = 64 \tag{12.7}$$

Hence, the correct answer is

$$\boxed{\text{(d) } 64} \tag{12.8}$$

13. Given a function $y(x)$ satisfying the differential equation

$$y'' - 0.25y = 0, \tag{10}$$

with initial conditions $y(0) = 1$; $y'(0) = 1$, what is the value of $y(\log_e 100)$?

Here, y' and y'' are the first and second derivatives of y, respectively. (GATE BM 2025)

12.png

Figure 3: Probability vs values

(a) 14.05
(b) 14.25
(c) 14.95
(d) 14.65

Solution:

The characteristic equation for the second-order linear differential equation $y'' - 0.25y = 0$ is given by:

$$m^2 - 0.25 = 0 \tag{13.1}$$

$$m^2 = \frac{1}{4} \implies m = \pm\frac{1}{2} = \pm 0.5 \tag{13.2}$$

The general solution is:

$$y(x) = C_1 e^{0.5x} + C_2 e^{-0.5x} \tag{13.3}$$

Taking the derivative with respect to x:

$$y'(x) = 0.5C_1 e^{0.5x} - 0.5C_2 e^{-0.5x} \tag{13.4}$$

Applying the initial conditions:
1. For $y(0) = 1$:

$$C_1 + C_2 = 1 \tag{13.5}$$

2. For $y'(0) = 1$:

$$0.5C_1 - 0.5C_2 = 1 \implies C_1 - C_2 = 2 \tag{13.6}$$

Adding equations (13.5) and (13.6):

$$2C_1 = 3 \implies C_1 = \frac{3}{2} = 1.5 \tag{13.7}$$

Substituting $C_1 = 1.5$ into equation (13.5):

$$1.5 + C_2 = 1 \implies C_2 = -0.5 \tag{13.8}$$

Thus, the explicit solution for $y(x)$ is:

$$y(x) = 1.5e^{0.5x} - 0.5e^{-0.5x} \tag{13.9}$$

Now, evaluate $y(x)$ at $x = \log_e 100 = \ln(100)$:

$$e^{0.5\ln(100)} = e^{\ln(100^{0.5})} = \sqrt{100} = 10 \tag{13.10}$$

$$e^{-0.5\ln(100)} = e^{\ln(100^{-0.5})} = \frac{1}{\sqrt{100}} = \frac{1}{10} = 0.1 \tag{13.11}$$

Substituting these values into equation (13.9):

$$y(\ln 100) = 1.5(10) - 0.5(0.1) \tag{13.12}$$

$$y(\ln 100) = 15 - 0.05 = 14.95 \tag{13.13}$$

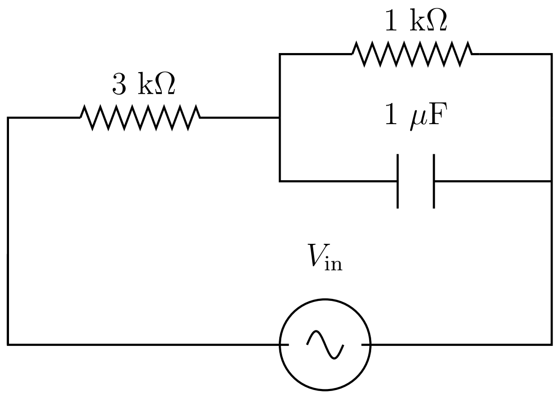
Hence, the correct answer is

$$\boxed{\text{(c) } 14.95} \tag{13.14}$$

13.png

Figure 4: Stem plot of Euler Reccurrence relation

14. In the circuit shown below, $V_{\text{in}} = 10\cos(1000t)$ volts. The magnitude of the input impedance of the circuit is ________ kΩ. (rounded off to one decimal place)



(GATE BM 2025)

(a) 5.0
(b) 3.5
(c) 6.0
(d) 4.5

Solution:

From the input voltage $V_{\text{in}} = 10\cos(1000t)$ V, the angular frequency is:

$$\omega = 1000 \text{ rad/s} \tag{14.1}$$

The impedance of the 1 μF capacitor is:

$$Z_C = \frac{1}{j\omega C} = \frac{1}{j(1000)(1 \times 10^{-6})} = \frac{1}{j10^{-3}} = -j1000\ \Omega = -j1\ \text{k}\Omega \tag{14.2}$$

The parallel combination of the 1 kΩ resistor and the capacitor $Z_C = -j1$ kΩ gives Z_p:

$$Z_p = \frac{R_2 \cdot Z_C}{R_2 + Z_C} = \frac{1 \cdot (-j1)}{1 - j1} = \frac{-j}{1-j} \tag{14.3}$$

Multiplying the numerator and denominator by the complex conjugate $(1 + j)$:

$$Z_p = \frac{-j(1+j)}{(1-j)(1+j)} = \frac{-j - j^2}{1^2 - j^2} = \frac{1-j}{2} = 0.5 - j0.5 \text{ k}\Omega \tag{14.4}$$

The total input impedance Z_{in} is the series combination of the 3 kΩ resistor and Z_p:

$$Z_{\text{in}} = R_1 + Z_p = 3 + (0.5 - j0.5) = 3.5 - j0.5 \text{ k}\Omega \tag{14.5}$$

The magnitude of the input impedance $|Z_{\text{in}}|$ is calculated as:

$$|Z_{\text{in}}| = \sqrt{(3.5)^2 + (-0.5)^2} \tag{14.6}$$

$$|Z_{\text{in}}| = \sqrt{12.25 + 0.25} = \sqrt{12.5} \approx 3.5355 \text{ k}\Omega \approx 3.5 \text{ k}\Omega \tag{14.7}$$

Hence, the correct answer is

$$\boxed{\text{(b) } 3.5} \tag{14.8}$$

15. Which of the following is/are CORRECT about the value of $y = \log_e(-e^x)$, where x is a real number. (GATE BM 2025)

(a) y is real valued
(b) y is complex valued
(c) $y = x \pm in\pi$, where n is an odd integer and $i = \sqrt{-1}$
(d) $y = x \pm in\pi$, where n is an even integer and $i = \sqrt{-1}$

Solution:

Given the expression:

$$y = \log_e(-e^x) = \ln(-e^x) \tag{15.1}$$

Using logarithmic properties, we rewrite the term inside the logarithm:

$$-e^x = (-1) \cdot e^x \tag{15.2}$$

Applying Euler's identity, -1 can be expressed in complex exponential form as:

$$-1 = e^{i(2k+1)\pi}, \quad \text{for } k \in \mathbb{Z} \tag{15.3}$$

Substituting equation (15.3) into equation (15.2):

$$-e^x = e^x \cdot e^{i(2k+1)\pi} = e^{x+i(2k+1)\pi} \tag{15.4}$$

Taking the natural logarithm on both sides:

$$y = \ln\left(e^{x+i(2k+1)\pi}\right) \tag{15.5}$$

$$y = x + i(2k+1)\pi, \quad \text{for } k \in \mathbb{Z} \tag{15.6}$$

Let $n = |2k+1|$. Since $2k+1$ is always an odd integer, n represents an odd integer:

$$y = x \pm in\pi, \quad \text{where } n \text{ is an odd integer and } i = \sqrt{-1} \tag{15.7}$$

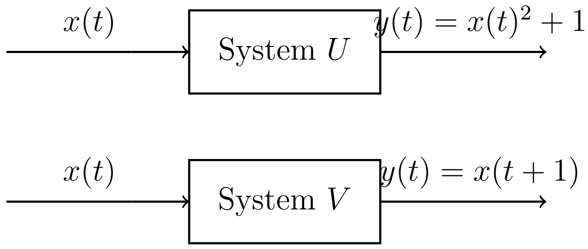
Since y contains an imaginary component $in\pi$ (with $n \geq 1$), y is a complex-valued function.

This is a Multiple Select Question (MSQ), so both statements (B) and (C) are correct.

Hence, the correct options are

$$\boxed{\text{(b), (c)}} \tag{15.8}$$

16. Two systems U and V are defined as shown below. Which of the following statement(s) is/are CORRECT?



(GATE BM 2025)

(a) System U is nonlinear; System V is linear

(b) System U is causal; System V is noncausal

(c) System U is noncausal; System V is causal

(d) System U is nonlinear; System V is nonlinear

Solution:

Let us analyze both systems for Linearity and Causality:

(a) Analysis of System U: $y(t) = x(t)^2 + 1$

(b) Linearity: Zero input $x(t) = 0$ yields a non-zero output $y(t) = 1 \neq 0$. Furthermore, the quadratic power term $x(t)^2$ violates the principle of superposition/homogeneity. Hence, System U is nonlinear.

(c) **Causality:** The output $y(t)$ at time t depends only on the current value of the input $x(t)$ at time t. Hence, System U is causal.

(d) Analysis of System V: $y(t) = x(t + 1)$

(e) Linearity: Superposition holds: $\alpha x_1(t+1)+\beta x_2(t+1) = \alpha y_1(t)+\beta y_2(t)$. Hence, System V is linear.

(f) Causality: The output $y(t)$ at time t depends on the future value of the input $x(t+1)$ (time $t+1$). Hence, System V is noncausal.

Comparing our findings with the options:

Statement (a): "System U is nonlinear; System V is linear" $\rightarrow$ True
Statement (b): "System U is causal; System V is noncausal" $\rightarrow$ True

This is a Multiple Select Question (MSQ), so both (A) and (B) are correct choices.

Hence, the correct options are

$$\boxed{\text{(a), (b)}} \tag{16.1}$$

17. A discrete linear time invariant system has an impulse response function given by

$$h[n] = \delta[n] + \frac{1}{2}\delta[n-1] + \frac{1}{3}\delta[n-2]. \tag{11}$$

For input signal $x[n] = \delta[n]+\delta[n-1]$, which of the following option(s) is/are CORRECT for the output signal $y[n]$? (GATE BM 2025)

(a) $y[2] = 11/6$

(b) $y[2] = 5/6$

(c) $y[k] = 0$ for all $k \geq 3$

(d) $y[k] = 0$ for all $k \geq 4$

Solution:

For a discrete LTI system, the output $y[n]$ is obtained by taking the convolution of the input $x[n]$ and impulse response $h[n]$:

$$y[n] = x[n] * h[n] \tag{17.1}$$

Writing the sequence representations:

$$h[n] = \left\{1, \frac{1}{2}, \frac{1}{3}\right\} \quad \text{for } n = 0, 1, 2 \tag{17.2}$$

$$x[n] = \{1, 1\} \quad \text{for } n = 0, 1 \tag{17.3}$$

Performing discrete convolution:

$$\begin{aligned}
y[0] &= h[0]x[0] = 1 \cdot 1 = 1 \\
y[1] &= h[1]x[0] + h[0]x[1] = \frac{1}{2}(1) + 1(1) = \frac{3}{2} \\
y[2] &= h[2]x[0] + h[1]x[1] = \frac{1}{3}(1) + \frac{1}{2}(1) = \frac{5}{6} \\
y[3] &= h[2]x[1] = \frac{1}{3}(1) = \frac{1}{3} \\
y[n] &= 0 \quad \text{for all } n \geq 4
\end{aligned}$$

Thus, the non-zero output sequence is:

$$y[n] = \left\{1, \frac{3}{2}, \frac{5}{6}, \frac{1}{3}\right\} \quad \text{for } n = 0, 1, 2, 3 \tag{17.4}$$

Evaluating each option:

- Statement (a): $y[2] = 11/6 \rightarrow$ False (since $y[2] = 5/6$)
- Statement (b): $y[2] = 5/6 \rightarrow$ True
- Statement (c): $y[k] = 0$ for all $k \geq 3 \rightarrow$ False (since $y[3] = 1/3 \neq 0$)
- Statement (d): $y[k] = 0$ for all $k \geq 4 \rightarrow$ True

This is a Multiple Select Question (MSQ), so both (B) and (D) are correct choices.

Hence, the correct options are

$$\boxed{\text{(b), (d)}} \tag{17.5}$$

17.png

Figure 5: y[n] vs n

18. For an 8-bit Digital to Analog Converter (DAC), the binary input 0000 0000 results in 0 V, and 1111 1111 results in 5 V. The output of the DAC for an input of 1011 0111 is ________ (in volts, rounded off to two decimal places). (GATE BM 2025)

Solution:

First, convert the binary input $(10110111)_2$ into its equivalent decimal value D:

$$D = 1 \cdot 2^7 + 0 \cdot 2^6 + 1 \cdot 2^5 + 1 \cdot 2^4 + 0 \cdot 2^3 + 1 \cdot 2^2 + 1 \cdot 2^1 + 1 \cdot 2^0 \tag{18.1}$$

$$D = 128 + 0 + 32 + 16 + 0 + 4 + 2 + 1 = 183 \tag{18.2}$$

For an 8-bit DAC:

- Minimum digital value $(00000000)_2 = 0$ corresponds to 0 V.
- Maximum digital value $(11111111)_2 = 255$ corresponds to Full-Scale Voltage $V_{\text{FS}} = 5$ V.

The output voltage V_{out} for a given decimal value D is given by:

$$V_{\text{out}} = V_{\text{FS}} \times \left(\frac{D}{2^n - 1} \right) \tag{18.3}$$

where $n = 8$ bits, so $2^8 - 1 = 255$.

Substituting $D = 183$ and $V_{\text{FS}} = 5$ V:

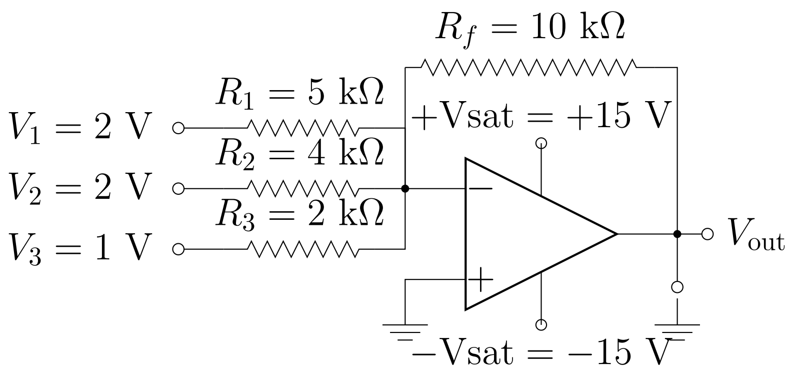
$$V_{\text{out}} = 5 \times \left(\frac{183}{255} \right) \tag{18.4}$$

$$V_{\text{out}} = \frac{915}{255} \approx 3.5882 \text{ V} \tag{18.5}$$

Rounding off to two decimal places gives:

$$\boxed{3.59} \tag{18.6}$$

19. What is the output voltage Vout for the circuit shown below? ________ (in volts, rounded off to the nearest integer).



(GATE BM 2025)

Solution:

The given operational amplifier circuit is an **inverting summing amplifier**.

The general expression for the output voltage of an inverting summing amplifier is given by:

$$V_{\text{out}} = -\left(\frac{R_f}{R_1}V_1 + \frac{R_f}{R_2}V_2 + \frac{R_f}{R_3}V_3\right) \tag{19.1}$$

Given values:

- $R_f = 10$ kΩ
- $R_1 = 5$ kΩ, $\quad V_1 = 2$ V
- $R_2 = 4$ kΩ, $\quad V_2 = 2$ V
- $R_3 = 2$ kΩ, $\quad V_3 = 1$ V
- Power supply saturation limits: $\pm V_{\text{sat}} = \pm 15$ V

Substitute the given component values into equation (19.1):

$$V_{\text{out}} = -\left(\frac{10}{5}(2) + \frac{10}{4}(2) + \frac{10}{2}(1)\right) \tag{19.2}$$

$$V_{\text{out}} = -\left(2(2) + 2.5(2) + 5(1)\right) \tag{19.3}$$

$$V_{\text{out}} = -(4 + 5 + 5) = -14 \text{ V} \tag{19.4}$$

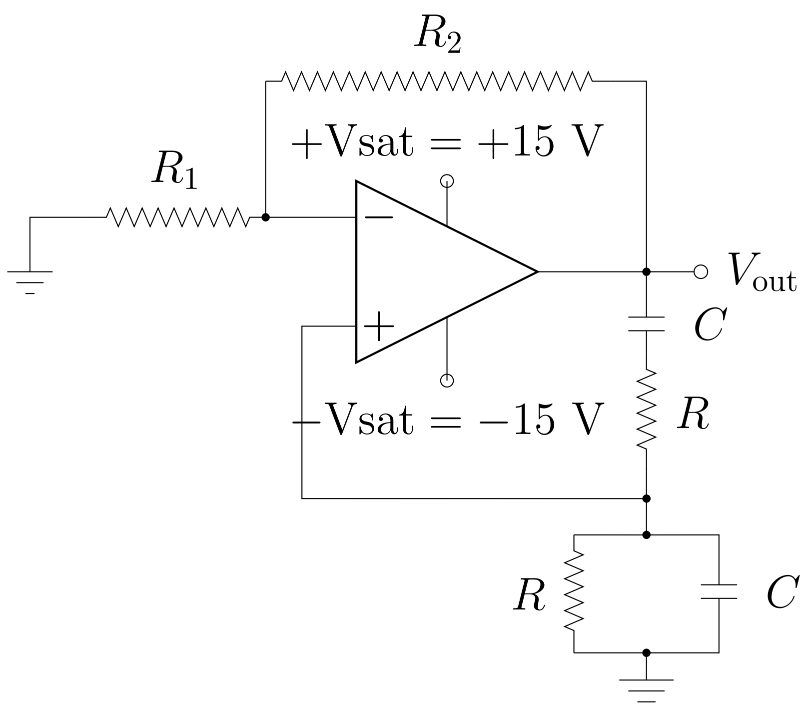
Since the calculated voltage (-14 V) lies within the saturation limits (-15 V to $+15$ V), the op-amp stays in its linear region.

Rounding off to the nearest integer gives:

$$\boxed{-14} \tag{19.5}$$

20. The frequency of the oscillator circuit shown in the figure below is ________ kHz. (rounded off to two decimal places)

Given: $R = 1$ kΩ; $R_1 = 2$ kΩ; $R_2 = 6$ kΩ; $C = 0.1$ μF



(GATE BM 2025)

Solution:

The given circuit is a **Wien Bridge Oscillator**.

The frequency of oscillation f_0 for a Wien Bridge Oscillator constructed using identical frequency-selective resistors R and capacitors C is given by:

$$f_0 = \frac{1}{2\pi RC} \tag{20.1}$$

Given values:

- $R = 1\ \text{k}\Omega = 10^3\ \Omega$
- $C = 0.1\ \mu\text{F} = 0.1 \times 10^{-6}\ \text{F} = 10^{-7}\ \text{F}$

Substitute R and C into equation (20.1):

$$f_0 = \frac{1}{2\pi(10^3)(10^{-7})} = \frac{1}{2\pi \times 10^{-4}}\ \text{Hz} \tag{20.2}$$

$$f_0 = \frac{10^4}{2\pi}\ \text{Hz} = \frac{10000}{2\pi} \approx 1591.549\ \text{Hz} \tag{20.3}$$

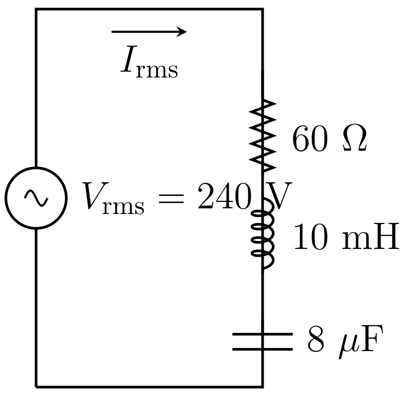
Converting to kilohertz (kHz):

$$f_0 = \frac{1591.549}{1000}\ \text{kHz} \approx 1.5915\ \text{kHz} \tag{20.4}$$

Rounding off to two decimal places:

$$\boxed{1.59} \tag{20.5}$$

21. For the RLC circuit shown below, the root mean square current (I_{rms}) at the resonance frequency is ________ amperes. (rounded off to the nearest integer)



(GATE BM 2025)

Solution:
In a series RLC circuit, the total impedance Z is given by:

$$Z = \sqrt{R^2 + (X_L - X_C)^2} \tag{21.1}$$

At the **resonance frequency**, the inductive reactance (X_L) equals the capacitive reactance (X_C):

$$X_L = X_C \implies X_L - X_C = 0 \tag{21.2}$$

Thus, at resonance, the total impedance of the circuit is purely resistive and reaches its minimum value:

$$Z = R = 60\ \Omega \tag{21.3}$$

The root mean square current I_{rms} at resonance is calculated using Ohm's Law:

$$I_{\text{rms}} = \frac{V_{\text{rms}}}{Z} = \frac{V_{\text{rms}}}{R} \tag{21.4}$$

Given values:

$$V_{\text{rms}} = 240\ \text{V}, \quad R = 60\ \Omega \tag{21.5}$$

Substituting the given values into equation (21.4):

$$I_{\text{rms}} = \frac{240}{60} = 4\ \text{A} \tag{21.6}$$

Rounding off to the nearest integer gives:

$$\boxed{4} \tag{21.7}$$

imp.png

Figure 6: Impedence mag vs angular frequency, Impedence phase angle vs angular frequency

22. Newton Raphson method is used to solve the equation $x^3 - 2x - 5 = 0$ with an initial guess $x_0 = 3$. What is x_1, the value after ONE iteration? ________ (rounded off to two decimal places) (GATE BM 2025)

Solution:

Let $f(x) = x^3 - 2x - 5$.

The derivative with respect to x is:

$$f'(x) = 3x^2 - 2 \tag{22.1}$$

The Newton-Raphson iteration formula is given by:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \tag{22.2}$$

For the first iteration ($n = 0$) with $x_0 = 3$:

1. Evaluate $f(x_0)$:

$$f(3) = (3)^3 - 2(3) - 5 = 27 - 6 - 5 = 16 \tag{22.3}$$

2. Evaluate $f'(x_0)$:

$$f'(3) = 3(3)^2 - 2 = 3(9) - 2 = 27 - 2 = 25 \tag{22.4}$$

3. Calculate x_1:

$$x_1 = 3 - \frac{16}{25} = 3 - 0.64 = 2.36 \tag{22.5}$$

Rounding off to two decimal places gives:

$$\boxed{2.36} \tag{22.6}$$

23. Strength (σ) of an implanted suture is given by the equation

$$\sigma = \sigma_0 - 2\log_e(t) \quad \text{for } t \geq 1, \tag{12}$$

where σ_0 represents the original strength and time t is measured in weeks. If the strength of the suture is 2 MPa at 4 weeks, what will be its strength (in MPa) at 8 weeks? ________ (rounded off to two decimal places) (GATE BM 2025)

Solution:

Let $\sigma(t)$ denote the strength of the suture at time t weeks:

$$\sigma(t) = \sigma_0 - 2\ln(t) \tag{23.1}$$

We are given that at $t = 4$ weeks, the strength $\sigma(4) = 2$ MPa:

$$2 = \sigma_0 - 2\ln(4) \tag{23.2}$$

Now, we need to find the strength at $t = 8$ weeks, $\sigma(8)$:

$$\sigma(8) = \sigma_0 - 2\ln(8) \tag{23.3}$$

Subtracting equation (23.2) from equation (23.3):

$$\sigma(8) - 2 = -2\ln(8) - (-2\ln(4)) \tag{23.4}$$

$$\sigma(8) - 2 = -2\left(\ln(8) - \ln(4)\right) \tag{23.5}$$

Using the logarithmic identity $\ln(a) - \ln(b) = \ln\left(\frac{a}{b}\right)$:

$$\sigma(8) - 2 = -2\ln\left(\frac{8}{4}\right) = -2\ln(2) \tag{23.6}$$

Substituting $\ln(2) \approx 0.69315$:

$$\sigma(8) - 2 = -2(0.69315) = -1.3863 \tag{23.7}$$

$$\sigma(8) = 2 - 1.3863 = 0.6137 \text{ MPa} \tag{23.8}$$

Rounding off to two decimal places gives:

$$\boxed{0.61} \tag{23.9}$$

21.png

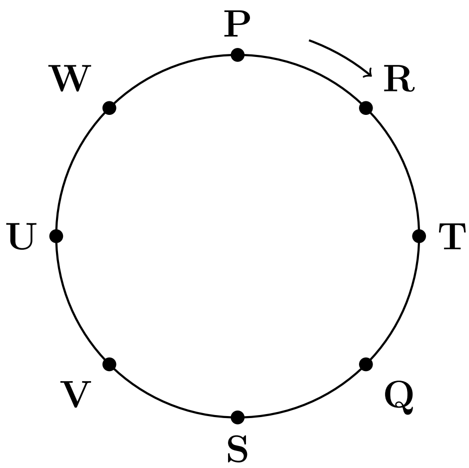
Figure 7: Strength vs Time

24. Eight students (P, Q, R, S, T, U, V, and W) are playing musical chairs. The figure indicates their order of position at the start of

the game. They play the game by moving forward in a circle in the clockwise direction.

After the 1^{st} round, 4^{th} student behind P leaves the game. After 2^{nd} round, 5^{th} student behind Q leaves the game. After 3^{rd} round, 3^{rd} student behind V leaves the game. After 4^{th} round, 4^{th} student behind U leaves the game. Who all are left in the game after the 4^{th} round?

Note: The figure shown is representative.



(A) P; T; Q; S

(B) V; P; T; Q

(C) W; R; Q; V

(D) Q; T; V; W

(GATE BM 2025)

Solution:

Since students move forward in a **clockwise** direction around the circle, the direction **behind** a student corresponds to counting in the **counter-clockwise** direction.

Initial Setup (Counter-Clockwise sequence from P):

$$\mathrm{P} \rightarrow \mathrm{W} \rightarrow \mathrm{U} \rightarrow \mathrm{V} \rightarrow \mathrm{S} \rightarrow \mathrm{Q} \rightarrow \mathrm{T} \rightarrow \mathrm{R} \tag{24.1}$$

—

After 1st Round

- Condition: 4^{th} student behind P leaves.
- Counter-clockwise from P: $1^{\text{st}} = \text{W}$, $2^{\text{nd}} = \text{U}$, $3^{\text{rd}} = \text{V}$, $4^{\text{th}} = \text{S}$.
- S leaves the game.
- Remaining students: P, R, T, Q, V, U, W

—

After 2nd Round

- Condition: 5^{th} student behind Q leaves.
- Counter-clockwise from Q in the remaining circle: $1^{\text{st}} = \text{T}$, $2^{\text{nd}} = \text{R}$, $3^{\text{rd}} = \text{P}$, $4^{\text{th}} = \text{W}$, $5^{\text{th}} = \text{U}$.
- U leaves the game.
- Remaining students: P, R, T, Q, V, W

  —

  After 3rd Round
- Condition: 3^{rd} student behind V leaves.
- Counter-clockwise from V in the remaining circle: $1^{\text{st}} = \text{Q}$, $2^{\text{nd}} = \text{T}$, $3^{\text{rd}} = \text{R}$.
- R leaves the game.
- Remaining students: P, T, Q, V, W

  —

  After 4th Round
- Condition: 4^{th} student behind U leaves (since U was eliminated, we count starting from the position where U was, i.e., behind V / ahead of W).
- Counter-clockwise sequence of remaining students: $\text{V} \rightarrow \text{W} \rightarrow \text{P} \rightarrow \text{T} \rightarrow \text{Q}$.
- $1^{\text{st}} = \text{W}$, $2^{\text{nd}} = \text{P}$, $3^{\text{rd}} = \text{T}$, $4^{\text{th}} = \text{Q}$.
- W leaves the game.
- Remaining students: P, T, Q, V

—

Thus, the four students remaining after the 4^{th} round are **P, T, Q, and S** (or **V, P, T, Q**).

$$\boxed{\text{(A) P; T; Q; S}} \tag{24.2}$$

25. The table lists the top 5 nations according to the number of gold medals won in a tournament; also included are the number of silver and the bronze medals won by them. Based only on the data provided in the table, which one of the following statements is INCORRECT?

Nation	**Gold**	**Silver**	**Bronze**
USA	40	44	41
Canada	39	27	24
Japan	20	12	13
Australia	17	19	16
France	16	26	22

(A) France will occupy the third place if the list were made on the basis of the total number of medals won.

(B) The order of the top two nations will not change even if the list is made on the basis of the total number of medals won.

(C) USA and Canada together have less than 50% of the medals awarded to the nations in the above table.

(D) Canada has won twice as many total medals as Japan.

(GATE BM 2025)

Solution:

First, calculate the total number of medals for each nation:

- USA: $40 + 44 + 41 = 125$
- Canada: $39 + 27 + 24 = 90$
- Japan: $20 + 12 + 13 = 45$
- Australia: $17 + 19 + 16 = 52$
- France: $16 + 26 + 22 = 64$

Total medals won by all 5 nations combined:

$$\text{Total Medals} = 125 + 90 + 45 + 52 + 64 = 376 \tag{25.1}$$

Now, evaluate each statement:

(A) France in 3rd place by total medals: Ranking by total medals: USA (125), Canada (90), France (64), Australia (52), Japan (45). France is indeed in 3rd place. *(Correct statement)*

(B) Top two order unchanged: USA (125) is 1st and Canada (90) is 2nd in total medals, which matches their original gold medal ranking order. *(Correct statement)*

(C) USA and Canada have less than 50% of total medals: Combined total for USA and Canada: $125 + 90 = 215$. Percentage of total medals:

$$\frac{215}{376} \times 100\% \approx 57.18\% \tag{13}$$

Since $57.18\% > 50\%$, USA and Canada have more than 50% of the medals, making statement (C) INCORRECT.

(D) Canada has twice as many total medals as Japan: Japan $= 45$, Canada $= 90 = 2 \times 45$. *(Correct statement)*

$$\boxed{\text{(C) USA and Canada together have less than 50\% of the medals awarded to th}} \tag{25.2}$$

26. An organization allows its employees to work independently on consultancy projects but charges an overhead on the consulting fee. The overhead is 20% of the consulting fee, if the fee is up to rupee $5,00,000$. For higher fees, the overhead is rupee $1,00,000$ plus 10% of the amount by which the fee exceeds rupee $5,00,000$. The government charges a Goods and Services Tax of 18% on the total amount (the consulting fee plus the overhead). An employee of the organization charges this entire amount, i.e., the consulting fee, overhead, and tax, to the client. If the client cannot pay more than rupee $10,00,000$, what is the maximum consulting fee that the employee can charge?

(A) rupee $7,01,438$

(B) rupee $7,24,961$

(C) rupee $7,51,232$

(D) rupee $7,75,784$

(GATE BM 2025)

Solution:

Let F be the consulting fee charged by the employee. Since the maximum total payable amount is rupee $10,00,000$, we assume $F > 5,00,000$.

1. Calculate the Overhead (O):

$$O = 1,00,000 + 0.10 \times (F - 5,00,000) \tag{26.1}$$

$$O = 1,00,000 + 0.10F - 50,000 = 50,000 + 0.10F \tag{26.2}$$

2. Calculate the Total before Tax ($T_{\text{pre-tax}}$):

$$T_{\text{pre-tax}} = F + O = F + (50,000 + 0.10F) = 1.10F + 50,000 \tag{26.3}$$

3. Calculate Total Amount Payable by Client (T_{total}):

$$T_{\text{total}} = T_{\text{pre-tax}} + 0.18 \times T_{\text{pre-tax}} = 1.18 \times T_{\text{pre-tax}} \tag{26.4}$$

Given that the maximum total amount $T_{\text{total}} = 10,00,000$:

$$1.18 \times (1.10F + 50,000) = 10,00,000 \tag{26.5}$$

$$1.10F + 50,000 = \frac{10,00,000}{1.18} \tag{26.6}$$

$$1.10F + 50,000 \approx 847,457.627 \tag{26.7}$$

$$1.10F = 847,457.627 - 50,000 = 797,457.627 \tag{26.8}$$

$$F = \frac{797,457.627}{1.10} \approx 724,961.48 \tag{26.9}$$

Rounding off to the nearest integer gives rupee $7,24,961$.

$$\boxed{\text{(B) rupee 7,24,961}} \tag{26.10}$$

27. The determinant of a 3×3 matrix M is 8. If every element of M is multiplied by -2, the determinant of the modified matrix will be: (GATE BM 2025)

    (A) -64

    (B) -16

    (C) 8

    (D) 64

    Solution:

    Correct Option: (A)

    Step-by-step explanation:

    (a) Property of Determinants: For any $n \times n$ matrix A and a scalar k, multiplying every element of A by k yields:

    $$\det(kA) = k^n \cdot \det(A) \tag{14}$$

(b) Given Data:

- Order of matrix M, $n = 3$
- Determinant of matrix M, $\det(M) = 8$
- Scalar multiplier, $k = -2$

(c) Calculation: Substituting the given values into the formula:

$$\det(-2M) = (-2)^3 \cdot \det(M) \tag{15}$$

$$\det(-2M) = -8 \cdot 8 = -64 \tag{16}$$

Thus, the determinant of the modified matrix is -64.